NASM Installation Guide for Linux (WSL, VirtualBox, VMware)

This guide explains how to install NASM (Netwide Assembler) on Linux. Students can choose to install Linux using WSL (Windows Subsystem for Linux), or run Linux in a virtual machine using VirtualBox or VMware. Popular Linux distributions such as Ubuntu or Kali Linux can be used.

Option 1 – Using WSL (Windows Subsystem for Linux)

This is ideal for Windows users who want to run Linux without a full virtual machine.

Step 1 – Enable WSL:

```
1. Open PowerShell as Administrator
   - Right-click the Start menu → Windows PowerShell (Admin) or Command Prompt (Admin)
2. Install WSL with your preferred Linux distribution:
   wsl --install
   OR
   wsl --install -d ubuntu
   OR
   wsl --install -d kali
3. Restart your computer when prompted
```

Step 2 – Setup and Update Linux:

```
1. Launch your installed Linux distribution from the Start menu
2. Create your Linux username and password
3. Update packages:
   sudo apt update && sudo apt upgrade -y
```

Step 3 – Install NASM:

```
sudo apt install nasm -y
nasm -v
```

Option 2 – Using VirtualBox or VMware

This method gives you a full Linux environment without affecting your main OS.

Step 1 – Download and Install a Virtual Machine Tool:

```
- VirtualBox: https://www.virtualbox.org/
- VMware Workstation Player: https://www.vmware.com/products/workstation-player.html
```

Step 2 – Get a Linux ISO:

```
- Ubuntu: https://ubuntu.com/download
- Kali Linux: https://www.kali.org/get-kali/
```

Step 3 – Create a New Virtual Machine:

```
1. Open VirtualBox or VMware → New VM
2. Select your downloaded ISO
3. Allocate at least 2GB RAM and 20GB disk space
4. Start the VM and follow the Linux installation process
```

Step 4 – Install NASM on Your Linux VM:

```
sudo apt update && sudo apt upgrade -y
sudo apt install nasm -y
nasm -v
```